

One-Page Project Report

Project Title

HieraSync: AI-Powered Organizational Workflow Management

Project Domain: Artificial Intelligence & Machine Learning (Workflow Automation)

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1. Problem Definition

Many organizations still depend on emails, spreadsheets, messaging applications, and manual processes to manage daily operations. As the organization grows, it becomes difficult to track tasks, approvals, responsibilities, and deadlines across different departments. Managers often face delayed communication, lack of transparency, missed deadlines, and inefficient workload distribution. Existing workflow management systems mainly focus on task assignment but do not provide intelligent decision support or predictive workflow management using Artificial Intelligence.

2. Need of the Project

Organizations require a centralized platform that can efficiently manage hierarchical workflows, task assignments, approvals, and communication. The system should automatically assign tasks according to organizational roles while allowing managers to monitor progress in real time. AI can improve productivity by identifying high-priority tasks, predicting delays, generating reminders, and providing performance insights. This helps organizations improve collaboration, reduce manual work, and make better decisions

3. Proposed Solution

HieraSync is an AI-powered workflow management system designed to automate organizational processes through intelligent task management and hierarchy-based workflow automation. The system consists of four main modules:

- **Organization Management:** Creates role-based organizational hierarchy and manages users.
- **Workflow & Task Management:** Assigns, tracks, and approves tasks with automated workflow.
- **AI Decision Support:** Prioritizes tasks, predicts delays, and sends smart reminders.
- **Analytics Dashboard:** Displays task progress, approvals, productivity, and performance reports.

4. Innovation / Uniqueness

Unlike traditional workflow management applications, HieraSync offers:

- AI-based task prioritization based on deadlines and workload.
- Hierarchy-driven workflow automation with role-based approvals.
- Predictive delay detection to identify tasks that may miss deadlines.
- Intelligent productivity analytics for employees and departments.
- Centralized platform for task management, approvals, communication, and reporting.

5. Expected Outcome

- Centralized workflow management system.
- Faster communication and collaboration across departments.
- Reduced manual work and paperwork.
- AI-generated reminders and intelligent task scheduling.
- Real-time monitoring of task progress and employee productivity.
- Improved transparency, accountability, and organizational efficiency.

6. Technologies / Tools Proposed

Programming Languages: Python, SQL, JavaScript

Frontend: Flutter

Backend: FastAPI (Python)

Database: PostgreSQL

AI & Machine Learning: Scikit-learn, Pandas, NumPy

Data Visualization: Plotly, Matplotlib

Development Tools: Visual Studio Code, Git & GitHub, Postman

7. Application Area

Colleges, Corporate Organizations, IT Companies, Government Organizations, Healthcare Organizations, Manufacturing Industries, Banking & Financial Services, Startups, and Small & Medium Enterprises (SMEs)..

8. References

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2. **Arun Kumar et al. (2026)**, Manage One AI: A Unified AI-Driven Productivity and Collaboration Platform for Intelligent Task and Workflow Management, International Research Journal on Advanced Engineering and Management (IRJAEM).
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4. **Abid Haleem, Mohd Javaid, Ravi Pratap Singh & Rajiv Suman (2025)**, Artificial Intelligence Applications for Business Process Management: A Systematic Review, Journal of Information Technology.
5. **Mohd Javaid et al. (2024)**, Applications of Artificial Intelligence for Enterprise Workflow Automation and Decision Support, Journal of Industrial Information Integration.

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